RV College of Engineering®

IDEA
InDustrial Engineering Association
R V College of Engineering

THE INSIGHT

A NEWSLETTER INITIATIVE FROM IDEA

2025

MAY - JUNE

Edition

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EDITOR'S *Note*



Dr. Rajeswara Rao KVS
Head of Department
IEM, RVCE



Dr. Ramaa A
Associate Dean - P&T,
Associate Professor,
IDEA Secretary, Dept. of IEM,
RVCE

Welcome to the IEM Department Newsletter!

Stay informed and connected with the latest updates on departmental events, research breakthroughs, student achievements, and faculty highlights. From academic milestones to innovations in teaching and learning, we bring you stories that reflect the vibrant life of the Industrial Engineering and Management community.

This newsletter also showcases extracurricular activities, including hackathons, technical fests, cultural events, and student club initiatives. We aim to foster collaboration, creativity, and leadership among students while providing a platform to explore interests beyond the classroom.

Let this newsletter be your gateway to discovery, connection, and continuous development in the world of IEM.



Disha Agarwal



S Trijal Kumar



Inaya J Gujaran

INDUSTRIAL ENGINEERS IN THE AGE OF AI

A DYNAMIC PARTNERSHIP

Industrial Engineers (IEs) are stepping into a transformative era and are becoming architects of intelligent systems in the AI era. where artificial intelligence (AI) is reshaping how systems are designed, optimized, and managed. Far from being replaced, IEs are becoming strategic enablers of AI-driven innovation across industries.

The key roles of IEs in the AI Era include:

- **Process Optimization & Intelligent Automation:** AI enables IEs to model, simulate, and optimize workflows using real-time data. Algorithms like genetic optimization and reinforcement learning help fine-tune production parameters. Example: AI-driven digital twins simulate entire factories to test changes before implementation.
- **Predictive Maintenance & Asset Reliability:** IEs deploy machine learning models to analyse sensor data (vibration, temperature, etc.) for early fault detection. Reduces downtime and maintenance costs by shifting from reactive to proactive strategies.
- **AI-Enhanced Quality Control:** Computer vision systems detect defects in real time with precision beyond human capability. Deep learning models classify anomalies and trigger corrective actions instantly.
- **Smart Supply Chain & Logistics:** IEs use AI for demand forecasting, route optimization, and inventory control. Reinforcement learning adapts logistics strategies to changing market conditions.
- **Human-Centered AI System Design:** IEs ensure AI systems are ergonomic, safe, and intuitive for human operators. Focus on human-AI collaboration rather than replacement designing interfaces that empower workers.

The tools and technologies IEs use in practice:

- **Machine Learning (ML):** Learns patterns from historical data to make predictions or decisions, Used for predictive maintenance, quality control, and process optimization. Example: ML models forecast equipment failures by analysing sensor data like vibration and temperature.
- **Deep Learning:** A subset of ML using neural networks with multiple layers. It powers computer vision for defect detection and speech recognition in human-machine interfaces. Example: Convolutional Neural Networks (CNNs) inspect products on assembly lines in real time.

- **Natural Language Processing (NLP):** Enables machines to understand and generate human language. NLP is used for automated documentation, multilingual support, and voice-controlled systems. Example: NLP helps engineers interact with AI systems using spoken commands or technical queries.
- **Robotics & Autonomous Systems:** AI-driven robots perform repetitive or hazardous tasks with precision. It is used in material handling, assembly, and inspection. Example: Autonomous robots sort and package goods in smart factories.
- **Digital Twins:** Virtual replicas of physical systems that simulate performance in real time. DTs are used for design validation, process optimization, and predictive analytics. Example: A digital twin of a production line helps test changes before physical implementation
- **Generative Design:** AI explores thousands of design options based on constraints and goals. It is used in product development, structural engineering, and sustainability. Example: Engineers use generative design to create lightweight, durable components

These technologies help improving efficiency through reducing waste, downtime, and manual effort. AI makes the system agile by adopting quickly to changing conditions. IEs can explore new designs and workflows by using innovation and automate hazardous tasks and improves ergonomics through AI.

Skills IEs must possess in AI Era

Apart from foundational knowledge & skills of IE and modelling capabilities, the following skills to be integrated to work in AI Era.

- **Data Science & AI Literacy:** Understanding algorithms, data pipelines, and model evaluation.
- **Systems Thinking:** Designing interconnected systems that adapt and learn.
- **Ethical & Responsible AI:** Ensuring fairness, transparency, and safety in AI deployment.
- **Cross-Functional Collaboration:** Working with data scientists, software engineers, and domain experts.

AI is not replacing Industrial Engineers, it's redefining their toolkit. The most successful IEs will be those who: think like systems architects; speak the language of data and design for adaptability and resilience. AI technologies are revolutionizing industrial engineering by enabling smarter, faster, and more adaptive systems.



Dr. K N Subramanya
Principal and Professor
RV College of Engineering, Bangalore

The new core team was selected through a structured process involving application screening and interviews by faculty and the outgoing team. Final selections were based on involvement, commitment, and vision for leading upcoming student activities.

Introducing CORE COMMITTEE 2025 -26



DISHA AGARWAL
President



NINGRAJ
Secretary



ABHISHEK
Finance & Sponsorship



SINCHAN RAI
Finance & Sponsorship



TRIJAL KUMAR
Media & Publicity



HEMAVATHI
Media & Publicity



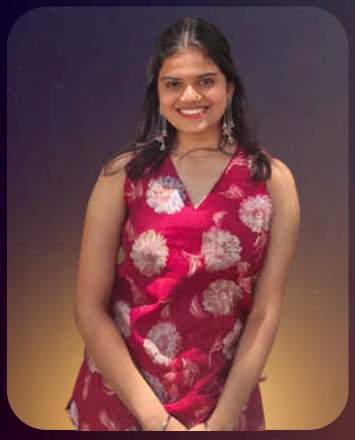
RAHUL ARUN
Events & Operation



SHIKSHA PANDEY
Events & Operation



MUSKAN SINGH
Academics & Placement



BHOOMIKA
Academics & Placement

Achievements



Nikitha Sunil of 6th Semester secured a summer internship at Schneider Electric. The Internship was related to productivity and quality functions of designing electrical switchgears.



Daivik R Aradhya of 6th Sem secured 4th place at the national-level Eurogrip Torq Challenge 3.0, held on May 9, 2025, by TVS Srichakra Ltd.



Mihika Kerur and **Adit Nahar of** 6th Sem secured 1st place in Pratidhwani – Battle of Bands at Utsav 2025, BMSCE's annual cultural fest.



Dheeraj Aradhya M, a 4th-semester IEM student, has earned the prestigious NCC 'B' Certificate, showcasing his dedication, discipline, and leadership.

STUDENT - FACULTY
Publications

The research paper titled "**Competitiveness Enablers in Handicraft Toy Manufacturing Industries Based on ISM-Fuzzy MICMAC Analysis**" by **V. Meghasree** and **Dr. C. K. Nagendra Guptha** has been published in IEEE Access (Volume 13) on 18 June 2025.

Scan to read

The Channapatna handicraft toy industry, known for its rich cultural heritage, faces challenges in staying competitive. A recent study using ISM and Fuzzy MICMAC analysis identified 11 key enablers; including marketing, branding, innovation, and government support as crucial for revitalizing the sector. The findings offer strategic insights to help stakeholders boost competitiveness and expand the industry's presence in domestic and global markets.

Authors : **Tanisha Kumari, Rishika Shreshtha, and Muskan Singh and their faculty mentor, Dr. Shobha N.S.,** Assistant Professor, Department of IEM, RVCE.



Scan to read

A research paper titled "Macroeconomic Models in Action: A Comparative Study of Keynesian, Classical, and Neo-Classical Theories" has been published in the prestigious International Journal of Creative Research Thoughts (IJCRT).

Authors: **Inchara N Aradhya, Suraj R S, R Chandra Kumar and Prof. Nandini B**, Department of IEM,RVCE.



The paper explores innovative methods for optimizing energy yield in wind farm projects. It is selected for publication in a Scopus-indexed journal.

The paper "Technology-Driven Methodology for Wind Resource Assessment and Energy Yield Optimization in Wind Farm Development" was presented at the 14th ICRCET, organized by IFERP.

Training





Dr. K. N. Subramanya - Principal, RVCE , **Dr. Rajeswara Rao K. V. S.** - HOD, IEM and **Prof. Bhaskar M. G.** - Faculty from the IEM Department, RVCE, conducted a 2-day Skill Up Program on AI and ML in Garment Manufacturing for ABFRL executives at ABFRL's Chandapura Plant on June 13–14, 2025.





Seminars / Conferences
Attended by Faculty

1. Dr. Ramaa A attended PayU sponsored Prosus Luminate event at Leela palace on 22nd May 2025

2. Dr. Vikram N Bahadurdesai attended “Deep Tech& Innovation Forum and Opening of the India Representation Office of Hof University in Bangalore” at Taj Hotel, Bangalore on 27th May 2025

3. **Dr. Rajeswara Rao K V S** attended the **BCIC Summit on AI in Manufacturing**, held at Lalit Ashok, Bengaluru, on June 4, 2025. The summit focused on the integration of artificial intelligence in manufacturing and its impact on industry practices.

4. **Dr. Ramaa A** participated in **Momentum’25**, an event organized by HirePro at Radisson Blu, Outer Ring Road, Bengaluru, on June 24, 2025.

5. Prof. Dr. Ramaa A, Prof. Dr. Shobha N S, Prof. Nandini B, Mr. N Govindarajulu, Mrs. Vidya V, completed training on POSH Act 2013 and UGC Regulations 2015, conducted by AICTE in an online session on Saturday, June 14, 2025. The training aimed at reinforcing awareness and compliance related to the Prevention of Sexual Harassment in academic institutions.

6. Dr. Vikram N B, MN Shruthi, Dinesh N attended an online training session on POSH (Prevention of Sexual Harassment) on June 21, 2025, conducted by Dr. Manoj R Wagh and organized by Tech Trainers and Testers, Pune.

7. Dr. Rajeswara Rao K V S, Dr. MN Vijayakumar, Naveen, attended “Prevention of Sexual Harassment course offered by NSDC, through Skill India Digital Hub(SIDH)” on 24thJune 2025.

FINAL YEAR
Major Project Exhibition

InDustrial Engineering Association

May - June 2025 EDITION

INDUSTRY EXPERTS - ALUMNI, IEM



Mr. Nishith Raj
Director Product Management
Zepto



Mr. Manoj K Y
Planning Analyst (Former)
Aditya Birla Fashion & Retail Ltd.



Ms. Revathi J
Business Transformation
Manager Accenture



Ms. Harshita U Kumar
Product Manager
TARGET

ACADEMIC EXPERTS



Dr. G.V. Prabhushankar
Professor and Head IEM, SIT,
Tumkuru



Dr. Rajeshwari P
Associate Prof. & Head
IEM, Dr. AIT



Dr. Shailaja V N
Professor & Head IEM,
BMSCE



Dr. Leela M H
Assistant Professor
Dept of MBA, Dr. AIT

BEST PROJECTS

Internal Projects

Sl. No.	Project Title	Place of work	Name of the students	Guide	Award
1.	Strategic EV Charging Station Allocation in Bengaluru Using Demand-Based Modeling and Metaheuristic Techniques	**RVCE,** *Bengaluru*	B N Vaishnavi H K Shreya Ujjwal B	**Dr.C K Nagendra Guptha** *Assoc. Prof & HoD, IEM, RVCE*	**First Best Project**
2.	Forecast-Driven Optimization of Karnataka's Energy Portfolio (2025–2030) Using Hard and Soft Computing Models	**RVCE,** *Bengaluru*	Aarohi Mahi Ayush Dudheria Ayush Singh Nilasish Roy	**Dr. N S Narahari** *Visiting Professor IEM, RVCE*	**Second Best Project**

Industry Projects

Sl. No.	Project Title	Place of work	Name of the students	Guide	Award
1.	Development of Standard Operating Procedures and Purchase Order Automation Framework for Wafer Fabrication Unit	**Global Foundries** *Bengaluru*	Gohitha M N Tanushri Ramana Gowda Vatsal S B	**Dr.Rajeswara Rao KVS** *Associate Professor IEM, RVCE*	**First Best Project**
2.	Development of Ikken Kanri Application for Assemblies Defects Management	**Toyota Kirloskar** *Auto Parts Pvt.Ltd. Bidadi*	Reeti Raj Shankaramma Sohilsab I	**Dr. K N Subramanya** *Professor & Principal IEM, RVCE*	**Second Best Project**
3.	Techno-Economic Feasibility Assessment for Bannihatti Wind Farm Project	**KREDL & IWPA** *Bengaluru*	Inchara N	**Prof. Nandini B** *Assistant prof. RVCE*	**Second Best Project**

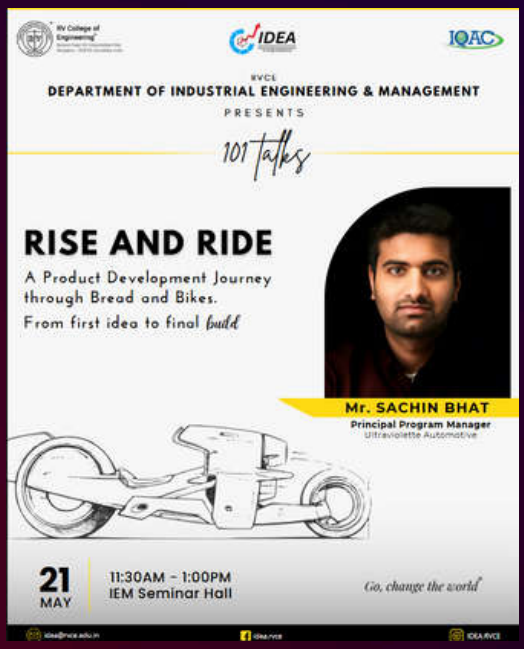




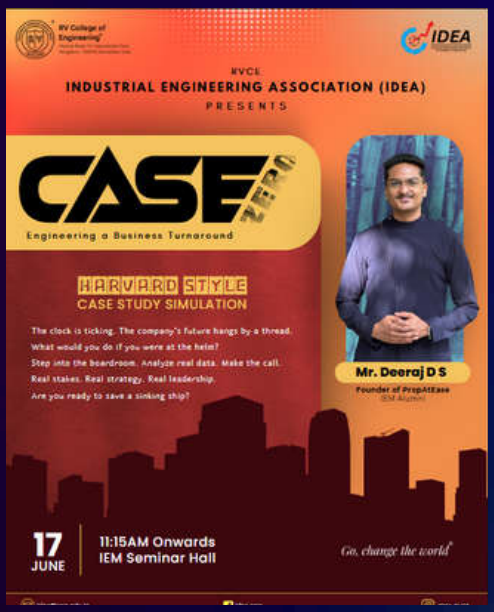




101 Talks



Mr. Sachin Bhat conducted a session on EV engineering roles, sharing insights from Tesla, Rivian, and Ultraviolette. His unique analogy between baking dough and product development made complex ideas relatable.



Mr. Deeraj D S led a crisis simulation titled "Case Zero – Air India Crash," where nine student teams acted as consulting firms. The session offered hands-on insights into corporate emergency handling.







Industrial visit to Polytec packins India pvt ltd,Harohalli on 26th May 2025

Our Graduating Batch – 2025

InDustrial Engineering Association

May - June 2025 EDITION





The Department of Industrial Engineering and Management hosted a heartfelt farewell for the Class of 2025 on June 5, organized by the IDEA Club with support from faculty and students.

The event featured cultural performances, heartfelt speeches, and reflections on the batch's journey. Professors shared words of wisdom, while students expressed gratitude, making it a joyful and memorable send-off. The department wishes the graduates great success in their future endeavors.



Major Events



On International Day of Yoga 2025, Dr. K. N. Subramanya - Principal, RVCE and NSS volunteers joined celebrations at Doordarshan Kendra, Bengaluru, themed “Yoga for Earth & One Health.” AICTE Chairman Prof. TG Sitharam was the guest of Honor for the event

The Department of Industrial Engineering and Management convened a Parent-Teacher Meeting to discuss students’ progress, attendance, and development. Parents appreciated the open communication and shared suggestions for continued growth and well-being.





59 students from the 6th semester RVCE (ME and IEM) successfully completed the Ready Engineer Program by Tata Technologies, gaining hands-on skills in CAD, design thinking, and industry practices. Their strong performance and enthusiasm reflect the department’s focus on practical, industry-oriented learning.

TATA TECHNOLOGIES
READY Engineer™

Berlin Talent Event

RVCE successfully hosted the Berlin Talent event, led by the Office of Global Partnerships, to promote international collaboration in innovation and entrepreneurship. The event was organized with active involvement from the IEM Department and the E-Cell.

Senator Franziska Giffey and delegates from 14 German companies engaged with students, strengthening Indo-German ties and expanding global opportunities.













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